CHEAT SHEET – MAC & HMAC (Stallings Style)

1. Core Concepts

A Message Authentication Code (MAC) is defined as T = MAC(K, M). It provides message integrity and authentication using a shared secret key, but does not provide confidentiality or nonrepudiation.

2. Hash vs MAC vs Digital Signature

1. Hash: Integrity only
2. MAC: Integrity + Authentication (shared key)
3. Digital Signature: Integrity + Authentication + Nonrepudiation

3. MAC Security Goal

The main security goal of a MAC is computational resistance: given known message–tag pairs, it should be infeasible to generate a valid MAC for any new message.

4. Existential Forgery

Existential forgery means that an attacker can produce at least one new message–MAC pair that is valid, without knowing the secret key, after observing other valid pairs. This leads to the standard notion of EUF-CMA security.

5. Brute-Force Security of MAC

1. Key-space attack: ~2^k operations
2. Tag-space attack: ~2^n operations
3. Effective security = min(2^k, 2^n)

6. HMAC – Standard Construction

HMAC uses a hash function as a black box and has provable security. It preserves hash performance, allows easy hash replacement, and provides simple key handling.

7. Hash Functions in Practice

1. Modern hashes: SHA-256, SHA-384, SHA-512, SHA-3
2. Legacy: MD5, SHA-1 (deprecated for standalone hashing)
3. Common HMACs: HMAC-SHA-256, HMAC-SHA-512